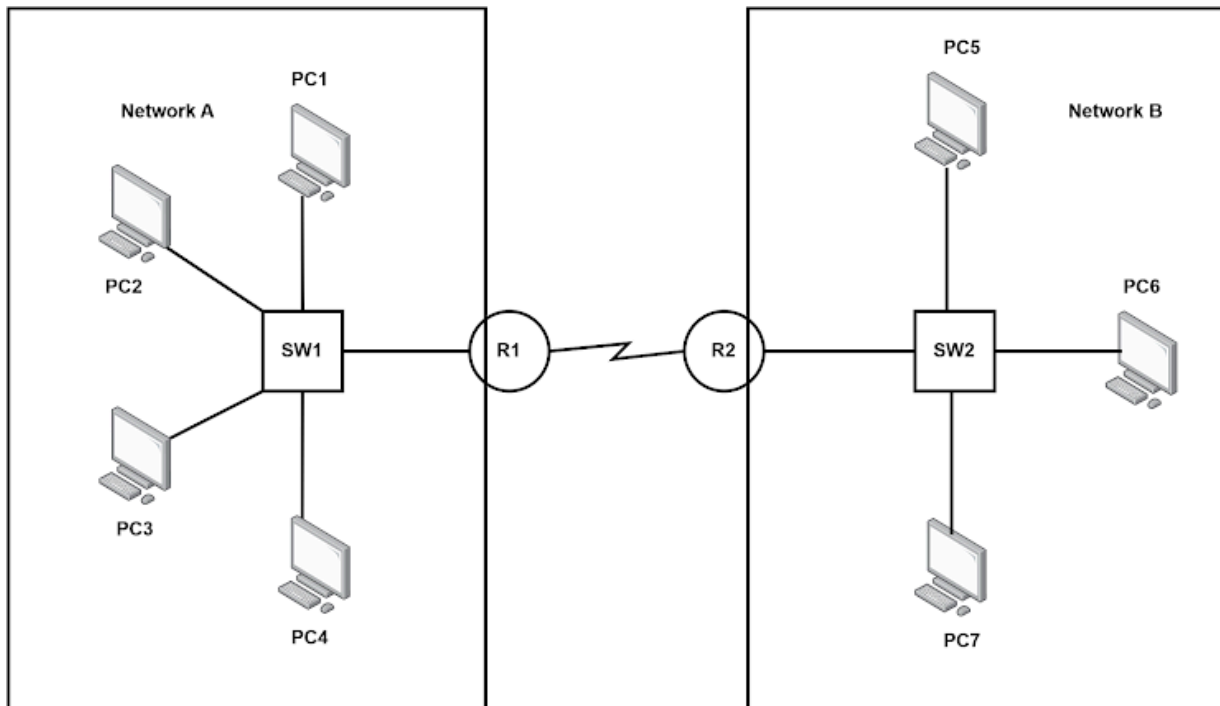


CSE421 Assignment 04

[MSMA | Summer 2025]

Answer all the questions in this form. You can submit only once even if you submit by mistake. Make sure that your answers are put correctly by refreshing the page.

Answer Q1 to Q7 based on the following scenario. IP address of PC3 is 172.168.10.194/12 and ip address of PC5 is 10.15.44.129/23



Q1. Calculate the subnet mask of Network B. (Format: 255.255.255.0)

Answer: 255.255.254.0

Q2. Calculate the network address of Network B. (Format: 10.10.10.10/24)

Answer: 10.15.44.0/23

Q3. Calculate the broadcast address of Network B. (Format: 10.10.10.10/24)

Answer: 10.15.45.255/23

Q4. Calculate the IP address of PC4 if it has the 1000th last usable address of the network. (Format: 10.10.10.10/24)

Answer: 172.175.252.23/12

Q5. Calculate the number of PCs that can be added more to network B (Format: 500)

Answer: 506

Q6. Calculate the destination address if PC3 wants to send a packet to all the devices of network B. (Format: 10.10.10.10)

Answer: 10.15.45.255

Q7. Calculate the destination address if PC3 wants to send a packet to all the devices of network A. (Format: 10.10.10.10)

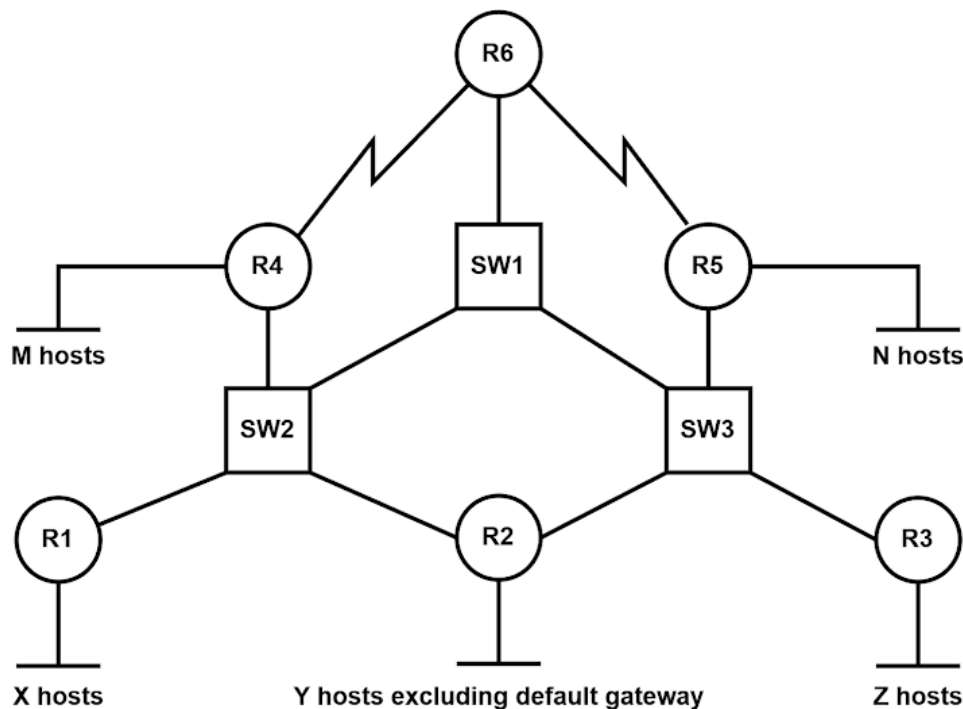
Answer: 255.255.255.255

Q8. What will be the simpler version for IPv6 address

0DAB:0010:0200:0002:A000:0000:0010:00B0?

Answer: DAB:10:200:2:A000::10:B0 or dab:10:200:2:a000::10:b0

Given the network address is 0.64.0.0/10, the value of M=4000, N=2000, X=1500, Y= 1022, Z=512, answer the following questions.



Q9. Calculate the total number of IP addresses that will be wasted if you use FLISM. [No relation with Q10]

Answer: 4185242

updated mask = 10+3 = 13. host bits for every network = 32-13 = 19

using 19 bits we get a total $2^{19} = 524288$ combinations/addresses for each network.

waste in switched network = $524288 - (7+2)$

waste in R1 LAN= $524288 - (1500+2)$

waste in R2 LAN= $524288 - (1022+1+2)$

waste in R3 LAN= $524288 - (512+2)$

waste in R4 LAN= $524288 - (4000+2)$

waste in R5 LAN= $524288 - (2000+2)$

waste in WAN network 1= $524288 - (2+2)$

waste in WAN network 2= $524288 - (2+2)$

add all these to find out the total waste

Q10. Calculate the maximum number of network addresses that you will get if you use FLSM given that the maximum requirement is based on the given topology.

Answer: 1024

Highest requirement 4000, total address needed 4002 so host bit needed 12. Since FLSM will have all networks with the same mask, we can't make any network having less than 12 host bits. so, number of new network bits that can be borrowed from host bits $22-12=10$. maximum networks = $2^{10} = 1024$

**Q11. Calculate the network address for the LAN connected to R1 using VLSM.
(Format: 10.10.10.10/24)**

Answer: 0.64.24.0/21

**Q12. Calculate the network address for the LAN connected to R2 using VLSM.
(Format: 10.10.10.10/24)**

Answer: 0.64.32.0/21

**Q13. Calculate the network address for the LAN connected to R3 using VLSM.
(Format: 10.10.10.10/24)**

Answer: 0.64.40.0/22

**Q14. Calculate the network address for the LAN connected to R4 using VLSM.
(Format: 10.10.10.10/24)**

Answer: 0.64.0.0/20

**Q15. Calculate the network address for the LAN connected to R5 using VLSM.
(Format: 10.10.10.10/24)**

Answer: 0.64.16.0/21

Q16. Calculate the network addresses of the rest of the networks if VLSM is used. (Format: 10.10.10.10/24, 11.11.11.11/24,)

Answer: 0.64.44.0/28, 0.64.44.16/30, 0.64.44.20/30

Q17. Calculate the total number of IP addresses that will be wasted if VLSM is used.

Answer: 2226

waste in switched network = $16 - (7+2)$

waste in R1 LAN= $2048 - (1500+2)$

waste in R2 LAN= $2048 - (1022+1+2)$

waste in R3 LAN= $1024 - (512+2)$

waste in R4 LAN= $4096 - (4000+2)$

waste in R5 LAN= $2048 - (2000+2)$

waste in WAN network 1= $4 - (2+2)$

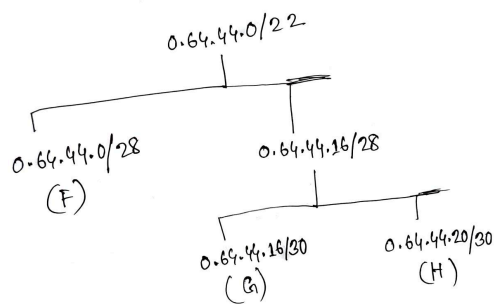
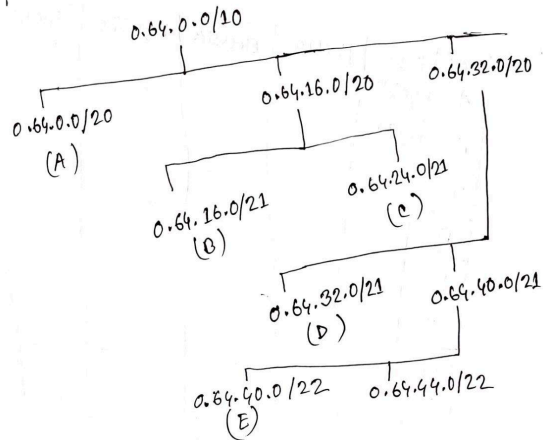
waste in WAN network 2= $4 - (2+2)$

add all these to find out the total waste

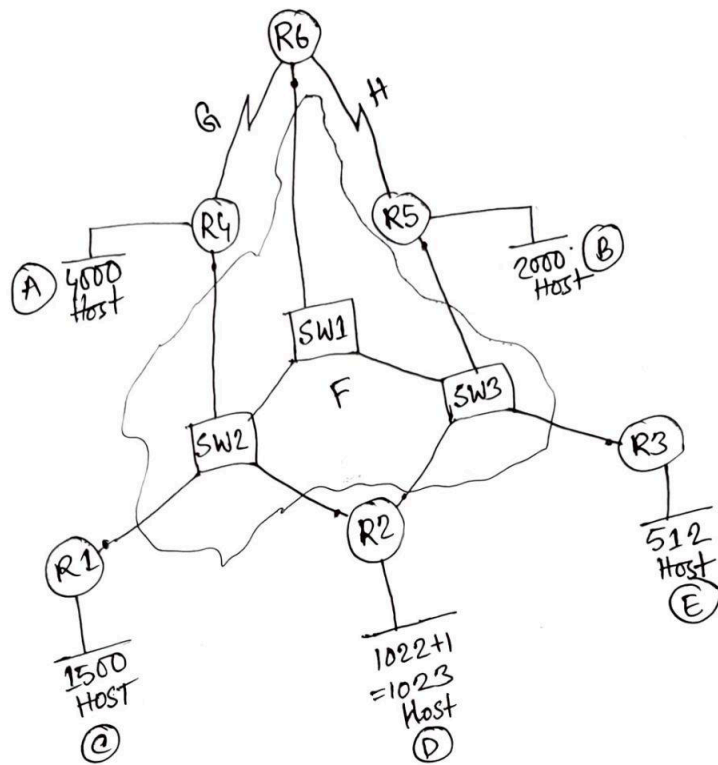
Q18. Submit Workings and Hierarchical Tree for VLSM.

0.64.0.0/10

00000000 00000000 00000000 → 0.64.0.0/20 ✓
 01 00000000 00000000 00000000 → 0.64.16.0/20 ✓
 01 00000000 00000000 00000000 → 0.64.16.0/21 ✓
 01 00000000 00000000 00000000 → 0.64.24.0/21 ✓
 01 00000000 00000000 00000000 → 0.64.32.0/21 ✓
 01 00000000 00000000 00000000 → 0.64.32.0/22 ✓
 01 00000000 00000000 00000000 → 0.64.40.0/22 ✓
 01 00000000 00000000 00000000 → 0.64.40.0/22 ✓
 01 00000000 00000000 00000000 → 0.64.44.0/22 ✓
 01 00000000 00000000 00000000 → 0.64.44.0/28 ✓
 01 00000000 00000000 00000000 → 0.64.44.16/28 ✓
 01 00000000 00000000 00000000 → 0.64.44.16/30 ✓
 01 00000000 00000000 00000000 → 0.64.44.20/30 ✓
 01 00000000 00000000 00000000 → 0.64.44.20/30 ✓



23719



Name	Host	Host+2	Block size	Host bits	Prefix
✓ A	4000	4002	4096	12	20
✓ B	2000	2002	2048	11	21
✓ C	1500	1502	2048	11	21
✓ D	1023	1025	2048	11	22
✓ E	512	514	1024	10	28
✓ F	7	9	16	4	30
✓ G	2	4	4	2	30
✓ H	2	4	4	2	